

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 2

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Goofy ah Math Number 2

Determine $\gcd(175, 63)$.

Theorem 3.2 [1] Euclidean Algorithm.

k	r_k	q_k	r_{k+1}
0	175	-	63
1	63	2	49
2	49	1	14
3	14	2	<u>7</u>

Table 1: Euclidean Algorithm finding $\gcd(175, 63)$

Determine the integers x and y so $7 = 175 \cdot x + 63 \cdot y$

Extended Euclidean algorithm - Bezouts identity: theorem 3.3 [1]

k	r_k	q_k	x_k	y_k
0	175	-	1	0
1	63	-	0	1
2	49	2	1	-2
3	14	1	-1	3
4	<u>7</u>	3	<u>4</u>	<u>-11</u>

Table 2: Extended Euclidean Algorithm finding integers x and y so $7 = 175 \cdot x + 63 \cdot y$

Convince God that there doesn't exist some integer r and s so $50 = 175 \cdot r + 63 \cdot s$

Corollary 3.5 [1]

For some numbers $a, b \in \mathbb{Z}$ that are not both zero, if $d = \gcd(a, b)$, then there is some integers r and s so that $d = ar + bs$ if and only if $d|c$.

This means there should exist some integer q such that $50 = 7q$. Sadly $\frac{50}{7} \notin \mathbb{Z}$ which means q doesn't exist and this follows that r and s don't exist either.

References

- [1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.